

Documentation for Ionization Update

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1 Analytic Solutions

- **Semi-Analytic Solution:** New advance_density updates in os-neutral.f03 and os-neutral-cyl-modes.f03 based on semi-analytic solution to the rate equations. The rate equations for a multi-level ionization are given by

$$\dot{N}_0 = -\omega_0 N_0 \quad (1)$$

$$\dot{N}_1 = -\omega_1 N_1 + \omega_0 N_0 \quad (2)$$

$$\vdots$$

where ω_j and N_j are the ionization rates and particle number for ionization level j , respectively. For this analysis, we assume ω_j are unique for each level j (e.g., $\omega_j \neq \omega_i$ for any $i \neq j$) and are considered constant during a single time-step dt . The solution for level 0 is trivial

$$N_0(t) = A_{0,0} \exp(-\omega_0 t). \quad (3)$$

where $A_0 = N_0(0)$. The next level can be solved using the ansatz

$$N_1(t) = A_{1,0} \exp(-\omega_0 t) + A_{1,1} \exp(-\omega_1 t) \quad (4)$$

and substituing into Eq. (2) to solve for the coefficients,

$$A_{1,0} = A_{0,0} \frac{\omega_0}{(\omega_1 - \omega_0)}, \quad A_{1,1} = N_1(0) - A_{1,0}. \quad (5)$$

Applying this methodology, the general solution for a given ionization level j

$$N_j(\Delta t) = \sum_{m=0}^j A_{j,m} \exp(-\omega_m \Delta t) \quad (6)$$

where

$$A_{j,m} = \begin{cases} A_{j-1,m} \times \omega_{j-1}/(\omega_j - \omega_m) & \text{if } m < j \\ N_j(0) - \sum_{k=0}^{m-1} A_{j,k} & m = j \end{cases} \quad (7)$$

2 Algorithm

- **Implementation:** The implementation of the recursive solution is as follows

Algorithm 1 Advance Density Algorithm

```

for each grid cell do
   $A(:, \text{exps}(\cdot)) \leftarrow 0, 0$ 
   $l \leftarrow 1$ 
  for  $m = 1 \dots \text{multi\_max}$  do
    if  $m > 1$  and  $\omega_m == 0$  and  $\omega_{m-1} == 0$  then
       $l \leftarrow m + 1$  ▷ Comment: reset recursion at m+1
      CYCLE
    end if
     $sum \leftarrow 0$ 
     $\text{exps}(m) \leftarrow e^{-\omega_m \Delta t}$ 
    for  $k = l \dots m - 1$  do
       $A_k \leftarrow A_k * \omega_{m-1}/(\omega_m - \omega_k)$ 
       $sum \leftarrow sum + A_k$ 
    end for
     $A_m \leftarrow N_m - sum$ 
     $N_m \leftarrow \sum_{j=l}^m A_j \text{exps}(j)$ 
  end for
end for

```

- **Run-time Complexity:** The analytic implementation has runtime $O(m^2)$ where m represents the the maximum ionization level. However, the existing approach in OSIRIS based on 2nd-order Runge-Kutta has runtime $O(m)$ and therefore remains the default algorithm.
- **Input Deck Setup:** The analytic algorithm can be toggled in the input deck by setting the **if_analytic** parameter in the neutral species. (see below)

```

!-----configuration of the Nitron 0-5-----
neutral
{
    neutral_gas = "N",
    multi_max = 7,

```

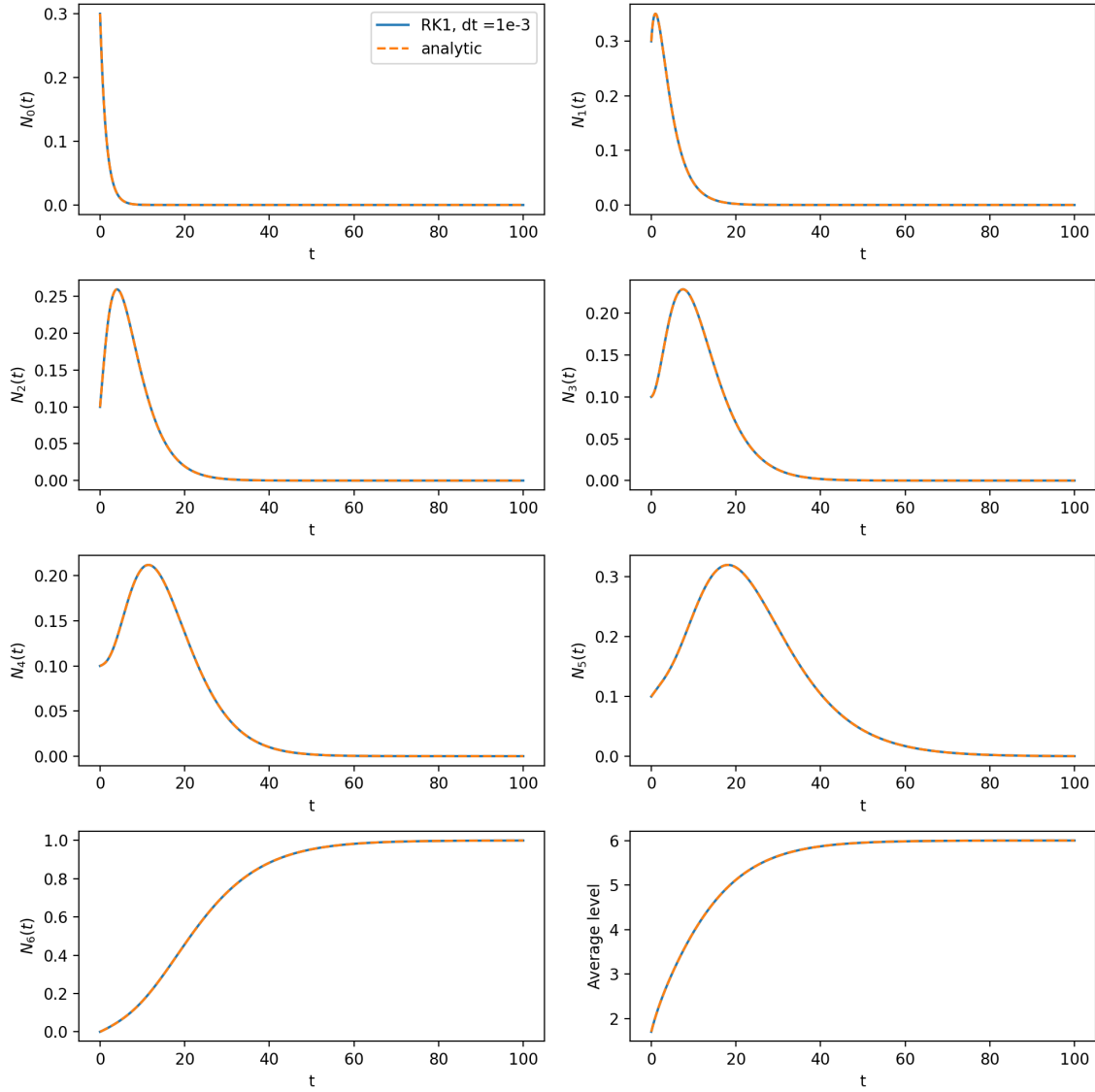
```

    multi_min = 5,
    if_analytic = .true.,
}

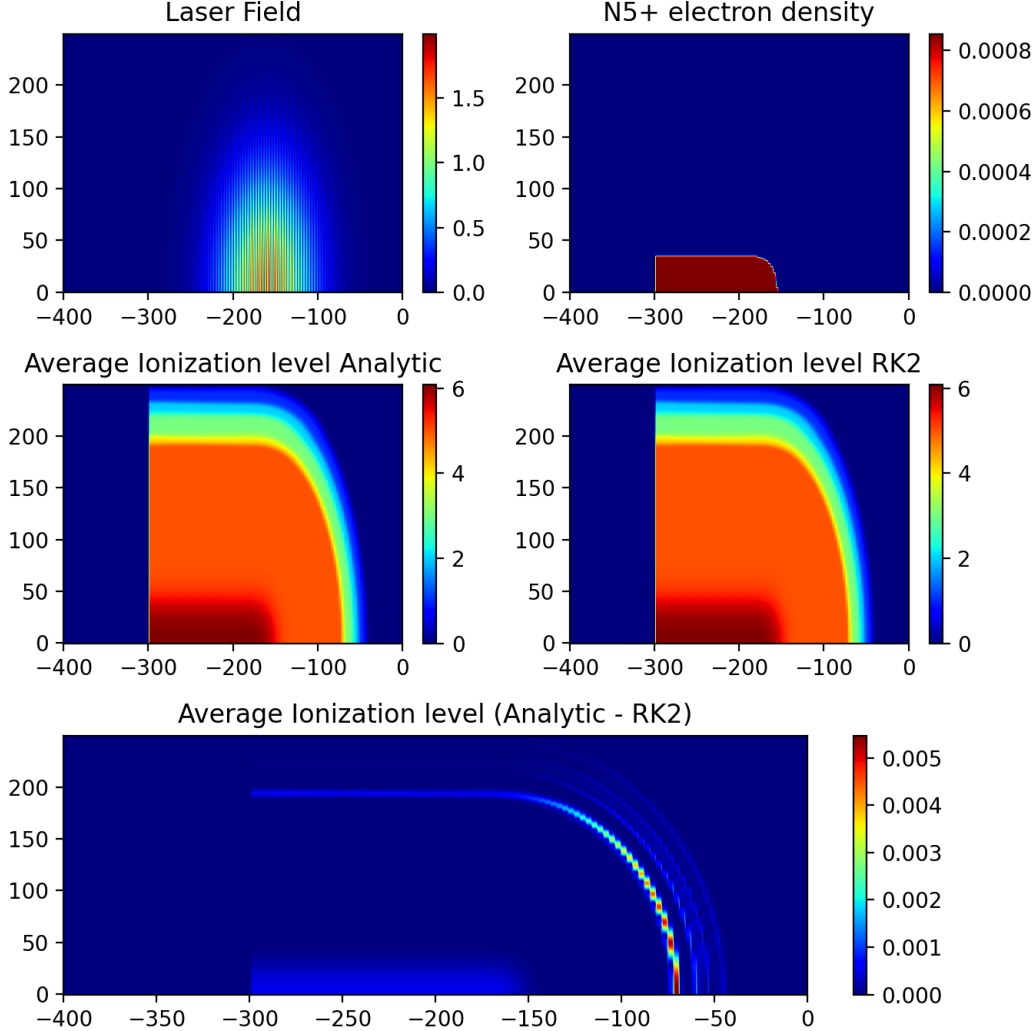
```

3 Testing

- **Comparisons of Offline RK1 and analytic algorithm:** The analytic solution reproduces results obtained using 1st order Runge-Kutta algorithm for multi-step ionization with a fine time step ($dt = 1e-3$). The results are shown below.



- **Quasi-3D Comparisons of RK2 and analytic algorithm:** Carried out quasi-3D LWFA simulations using the default RK2 algorithm in OSIRIS and the analytic algorithm. Strong qualitative agreement is observed when comparing the average ionization rates (below). The total simulation run time for the analytic algorithm is approximately 10% longer than the RK2 algorithm.



4 Miscellaneous

- **Minor Bug Fixes:** Bug fix in RK2 algorithm where cons is now $N(0) \frac{\omega \Delta t}{(1+0.5\omega \Delta t)}$ instead of $N(0)\omega \Delta t(1+0.5\omega \Delta t)$. Additional bug fix in `move_window_neutral()` is `os-neutral.f03` to account for guard cells when setting neutral density.